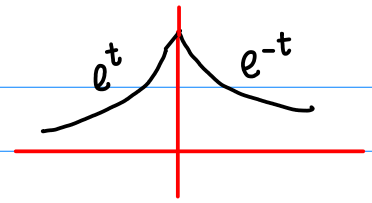
15|10|2020

L20 $\quad \mathcal{L}_2 \,,\ \mathcal{L}_{2+} \,,\ \mathcal{L}_{2-} \longleftarrow$ time domain

$\mathcal{L}_2$



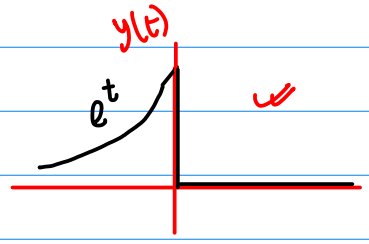
$$\int_{-\infty}^{\infty} e^{-|t|}\,dt = \int_{-\infty}^{0} e^{t}\,dt + \int_{0}^{\infty} e^{-t}\,dt$$

$$= 2\int_{0}^{\infty} e^{-t}\,dt < \infty$$

✓ $\mathcal{L}_{2+}$

$$\langle x(t), y(t)\rangle = \int_{-\infty}^{\infty} x^{*}(t)\, y(t)\,dt = 0$$

✓ $\mathcal{L}_{2-}$



Let $\mathcal{H}$ be a Hilbert space and $\mathcal{M} \subseteq \mathcal{H}$. Then the orthogonal complement of $\mathcal{M}$ is defined as —

$$\mathcal{M}^{\perp} = \{x \mid \langle x, y\rangle = 0 \ \ \forall\ y \in \mathcal{M},\ x \in \mathcal{H}\}$$

$\mathcal{M}^{\perp}$ is also Hilbert space.

e.g. $\mathbb{R}^2$ $\quad \mathcal{M} = \left\langle \begin{pmatrix}1\\1\end{pmatrix} \right\rangle \subseteq \mathbb{R}^2 \quad \mathcal{M}^{\perp} = \left\langle \begin{pmatrix}1\\-1\end{pmatrix} \right\rangle = \left\langle \begin{pmatrix}1\\1\end{pmatrix} \right\rangle^{\perp}$

$\langle x, y\rangle = x^{*}y$

$$\begin{cases} \alpha\begin{bmatrix}1\\1\end{bmatrix} = v \in \mathcal{M} \\ \beta\begin{bmatrix}1\\-1\end{bmatrix} = w \in \mathcal{M}^{\perp} \\ \langle v, w\rangle = v^{*}w = 0 \end{cases}$$

e.g. $\mathcal{L}_2$ $\quad \boxed{\mathcal{L}_{2-}^{\perp} = \mathcal{L}_{2+}}$

Let $\mathcal{M}$ and $\mathcal{N}$ be subspaces of $\mathcal{V}$. $\mathcal{V}$ is said to be a direct sum of $\mathcal{M}$ and $\mathcal{N}$ if $\mathcal{M} \cap \mathcal{N} = \{0\}$ and every element $x \in \mathcal{V}$ can be uniquely written as $x = m + n$, where $m \in \mathcal{M}$ and $n \in \mathcal{N}$.

$\begin{pmatrix}1\\0\end{pmatrix}$

e.g. $\mathbb{R}^2$ $\quad \mathcal{M} = \left\langle \begin{pmatrix}1\\1\end{pmatrix} \right\rangle,\ \mathcal{N} = \left\langle \begin{pmatrix}1\\-1\end{pmatrix} \right\rangle \qquad \begin{pmatrix}1\\0\end{pmatrix} = \frac{1}{2}\begin{pmatrix}1\\1\end{pmatrix} + \frac{1}{2}\begin{pmatrix}1\\-1\end{pmatrix}, \quad \left\langle \begin{pmatrix}1\\1\end{pmatrix} \right\rangle \cap \left\langle \begin{pmatrix}1\\-1\end{pmatrix} \right\rangle = \phi$

$\mathbb{R}^2 = \left\langle \begin{pmatrix}1\\1\end{pmatrix} \right\rangle \oplus \left\langle \begin{pmatrix}1\\-1\end{pmatrix} \right\rangle$

$\mathbb{R}^2$ $\quad \mathcal{M} = \left\langle \begin{pmatrix}1\\0\end{pmatrix} \right\rangle, \quad \mathcal{N} = \left\langle \begin{pmatrix}1\\1\end{pmatrix} \right\rangle \quad \boxed{\mathcal{M} \neq \mathcal{N}^\perp}$

$$\begin{pmatrix}1\\2\end{pmatrix} = -1\begin{pmatrix}1\\0\end{pmatrix} + 2\begin{pmatrix}1\\1\end{pmatrix} \qquad \mathcal{M} \cap \mathcal{N} = \phi \qquad \mathbb{R}^2 = \mathcal{M} \oplus \mathcal{N}$$

$$\mathcal{M} = \left\langle \begin{pmatrix}1\\0\end{pmatrix} \right\rangle \quad \boxed{\mathcal{N} = \left\langle \begin{pmatrix}1\\1\end{pmatrix} \right\rangle} \quad \mathcal{K} = \left\langle \begin{pmatrix}0\\1\end{pmatrix} \right\rangle$$

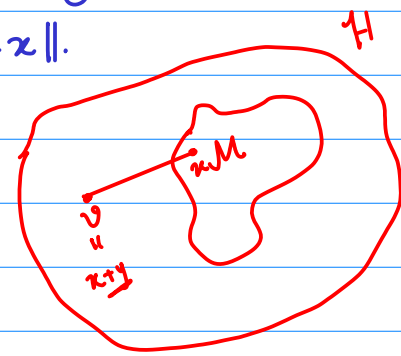
$$\mathbb{R}^2 = \mathcal{M} + \mathcal{N} + \mathcal{K} \qquad \begin{pmatrix}1\\2\end{pmatrix} = 1\begin{pmatrix}1\\0\end{pmatrix} + 2\begin{pmatrix}0\\1\end{pmatrix} = -1\begin{pmatrix}1\\0\end{pmatrix} + 2\begin{pmatrix}1\\1\end{pmatrix}$$

e.g. $\quad \ell_2 = \ell_{2-} \oplus \ell_{2+}$ (orthogonal direct sum)

$$\ell_2(-\infty, \infty) = \ell_2(0, \infty) \oplus \ell_2(-\infty, 0)$$

Closed subspace: $S \subseteq \mathcal{V}$ $\quad$ Every sequence in S that converges in $\mathcal{V}$ needs to converge in S.

Orthogonal projection theorem: Let $\mathcal{H}$ be a Hilbert space and $\mathcal{M}$ be a closed subspace of $\mathcal{H}$. Then for each vector $v \in \mathcal{H}$, $\exists$ a unique vector $x \in \mathcal{M}$ and $y \in \mathcal{M}^\perp$ such that $v = x + y$, i.e. $\mathcal{H} = \mathcal{M} \oplus \mathcal{M}^\perp$. Moreover $x \in \mathcal{M}$ is the unique vector s.t. $d(v, \mathcal{M}) = \|v - x\|$.



$\mathbb{R}^2$: $v = \begin{pmatrix}1\\2\end{pmatrix} \qquad \mathcal{M} = \left\langle \begin{pmatrix}1\\0\end{pmatrix} \right\rangle \qquad \mathcal{M}^\perp = \left\langle \begin{pmatrix}0\\1\end{pmatrix} \right\rangle$

$$\underset{v}{\begin{pmatrix}1\\2\end{pmatrix}} = 1\underset{x}{\begin{pmatrix}1\\0\end{pmatrix}} + 2\underset{y}{\begin{pmatrix}0\\1\end{pmatrix}} \qquad d(v, \mathcal{M}) = \left\| \begin{pmatrix}1\\2\end{pmatrix} - \begin{pmatrix}1\\0\end{pmatrix} \right\|$$

$$\mathcal{M} = \left\langle \begin{pmatrix}1\\1\end{pmatrix} \right\rangle \qquad \mathcal{M}^\perp = \left\langle \begin{pmatrix}1\\-1\end{pmatrix} \right\rangle \qquad \underset{v}{\begin{pmatrix}1\\2\end{pmatrix}} = \underset{x}{\tfrac{1}{2}\begin{pmatrix}1\\1\end{pmatrix}} + \underset{y}{\tfrac{1}{2}\begin{pmatrix}1\\-1\end{pmatrix}}$$

$$d(v, \mathcal{M}) = \left\| \begin{pmatrix}1\\2\end{pmatrix} - \begin{pmatrix}1/2\\1/2\end{pmatrix} \right\|$$

Bounded linear operators:

Suppose $\mathcal{V}$ and $\mathcal{Z}$ are Banach spaces. A mapping $F: \mathcal{V} \longrightarrow \mathcal{Z}$ is called bounded linear operator if

① $F(\alpha x + \beta y) = \alpha F(x) + \beta F(y)$, $\quad x, y \in \mathcal{V}$ and $\alpha, \beta \in \mathbb{F}$

(ii) $\exists$ a scalar $\alpha \geq 0$ such that

$$\|Fv\|_Z \leq \alpha \|v\|_V \quad \forall \; v \in V$$

The space of bounded linear operator is $\mathcal{BL}(V, Z)$, $\mathcal{L}(V, Z)$, $\mathcal{B}(V, Z)$

Induced norm: $\|F\|_{V \to Z} = \sup\limits_{\substack{v \in V \\ v \neq 0}} \dfrac{\|Fv\|_Z}{\|v\|_V}$

$\mathcal{BL}$ are complete

Banach space

e.g. $A : \mathbb{C}^n \longrightarrow \mathbb{C}^m$

$\|Av\|_2 \leq \|A\|_2 \|v\|_2$

$\leq \sigma_1 \|v\|_2$

σ_1 = highest singular value

$\|Av\|_p < \alpha \|v\|_p$

$v \in \mathbb{C}^n$

In finite dim^n spaces:

all p-norms are eq.

$\|x\|_1 \leq \|y\|_1$

$\Rightarrow \|x\|_p \leq \|y\|_p$